

System Model Formulas

Optimizing ABR Video Streaming in SDN-VANET-UAV using D3QN

Thesis Reference

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1 Communication Model

1.1 Path Loss: LoS and NLoS

The average path loss (dB) between UAV l and user k is computed as a weighted sum of LoS and NLoS components [1]:

$$g_{l,k} = P_{l,k}^{\text{LoS}} \cdot g_{l,k}^{\text{LoS}} + (1 - P_{l,k}^{\text{LoS}}) \cdot g_{l,k}^{\text{NLoS}} \quad (\text{Xie 1-2})$$

where the LoS and NLoS path loss terms are:

$$g_{l,k}^{\text{LoS}} = 20 \log_{10} \left(\frac{4\pi f_c d_0}{c} \right) + 10 n_{\text{LoS}} \log_{10}(d_{l,k}) + s_{\text{LoS}} \quad (\text{Xie 1})$$

$$g_{l,k}^{\text{NLoS}} = 20 \log_{10} \left(\frac{4\pi f_c d_0}{c} \right) + 10 n_{\text{NLoS}} \log_{10}(d_{l,k}) + s_{\text{NLoS}} \quad (\text{Xie 2})$$

Variables: f_c = carrier frequency; d_0 = reference distance; c = speed of light; $n_{\text{LoS}}, n_{\text{NLoS}}$ = path loss exponents; $s_{\text{LoS}}, s_{\text{NLoS}}$ = shadowing factors (dB); $d_{l,k} = \sqrt{(X_l - X_k)^2 + (Y_l - Y_k)^2 + H^2}$ = 3D distance UAV–user.

1.2 LoS Probability

$$P_{l,k}^{\text{LoS}} = \frac{1}{1 + \kappa \exp(-\zeta (\theta_{l,k} - \kappa))} \quad (\text{Xie 3})$$

where $\theta_{l,k} = \arcsin(H/d_{l,k})$ is the elevation angle (degrees), and κ, ζ are environment-dependent constants.

1.3 SINR: UAV $\rightarrow$ User (Downlink)

$$\text{SINR}_{l,k} = \frac{P_{\text{UAV}} \cdot 10^{-g_{l,k}/10}}{\sum_{l' \in \mathcal{L}, l' \neq l} P_{\text{UAV}} \cdot 10^{-g_{l',k}/10} + \sigma^2} \quad (\text{Xie 4})$$

Variables: P_{UAV} = UAV transmit power (W); σ^2 = noise power; interference from all other UAVs $l' \neq l$.

1.4 Downlink Rate: UAV $\rightarrow$ User

$$r_{l,k} = \frac{B}{M} \log_2(1 + \text{SINR}_{l,k}) \quad (\text{Xie 5})$$

Variables: B = total bandwidth; M = number of users / sub-channels per UAV.

1.5 Backhaul: MBS $\rightarrow$ UAV [Xie Eq. (6)–(9)]

In [1], the LoS and NLoS path losses for the wireless backhaul link (notation as in the paper) are

$$g_{\text{BS},l}^{\text{LoS}} = d_{\text{BS},l}^{-\gamma}, \quad (\text{Xie 6})$$

$$g_{\text{BS},l}^{\text{NLoS}} = \eta d_{\text{BS},l}^{-\gamma}. \quad (\text{Xie 7})$$

The average path loss is $g_{\text{BS},l} = P_{\text{BS},l}^{\text{LoS}} g_{\text{BS},l}^{\text{LoS}} + (1 - P_{\text{BS},l}^{\text{LoS}}) g_{\text{BS},l}^{\text{NLoS}}$ (see the text immediately after (7) in [1]). The received SINR is

$$\text{SINR}_{\text{BS},l} = \frac{P_{\text{BS}} 10^{-g_{\text{BS},l}/10}}{\sigma^2} \quad (\text{Xie 8})$$

and the backhaul rate is

$$r_{\text{BS},l} = \frac{B_h}{L} \log_2(1 + \text{SINR}_{\text{BS},l}). \quad (\text{Xie 9})$$

Variables: B_h = backhaul bandwidth; L = number of UAVs; P_{BS} = MBS transmit power; σ^2 = noise power; γ, η = backhaul path loss exponent and NLoS excess factor as in [1].

Code note: `models._rate_backhaul` computes an equivalent weighted channel gain (LoS probability $\times$ distance-dependent term) then SINR and $r_{\text{BS},l}$; the structure matches (8)–(9), while the split (6)/(7) is folded into one gain expression for implementation.

2 Request indicator, popularity, and cache (Xie)

Theo [1] (Section III-A2): Ký hiệu $p_{k,f,z} \in \{0, 1\}$ là *chỉ báo*: $p_{k,f,z} = 1$ khi user k đang yêu cầu chunk $v_{f,z}$, ngược lại bằng 0. Các yêu cầu chunk chịu ảnh hưởng độ phổ biến; trong bài báo, “*the popularity of video chunks follows a Zipf-like distribution*” [1] (tham chiếu [37] trong bài gốc). Các Eq. (Xie 10)–(Xie 12) lấy tổng theo (f, z) với các hệ số $p_{k,f,z}$; với mỗi request thực tế chỉ một cặp (f, z) có $p_{k,f,z} = 1$, các hạng tử còn lại bằng 0.

Cài đặt mô phỏng (luận văn): Mỗi bước cần *một* cặp (f, z) theo độ phổ biến Zipf-like. Code dùng phân phối xác suất joint trên tất cả cặp (f, z) với trọng số Zipf theo thứ hạng $r = 1, \dots, FZ$ (`environment._compute_zipf_joint_probs`), rồi lấy mẫu một cặp — tương đương việc sinh một mẫu chỉ báo $p_{k,f,z}$ trong mô phỏng Monte Carlo. Chi tiết công thức Zipf trên từng cặp là *lựa chọn implementation*; bài Xie chỉ nêu Zipf-like, không ghi đủ hệ số từng rank.

2.1 Cache placement and capacity [Xie, constraint (15a)]

Ma trận cache nhị phân $y_{l,f,z} \in \{0, 1\}$ (UAV l lưu chunk f ở bitrate z).
Ràng buộc dung lượng:

$$\sum_{f=1}^F \sum_{z=1}^Z y_{l,f,z} s_{f,z} \leq C_l^{\text{cache}}, \quad \forall l \in \mathcal{L}. \quad (\text{Xie 15a})$$

3 Video Delivery Delay Model

3.1 Total Delivery Delay

$$D_{l,k} = D_{l,k}^1 + D_{l,k}^2 + D_{l,k}^3 \quad (\text{Xie 13})$$

3.2 Scenario 1 — Direct Cache Hit

UAV l caches content f at exactly the requested bitrate z .

$$D_{l,k}^1 = \sum_{f=1}^F \sum_{z=1}^Z p_{k,f,z} y_{l,f,z} \frac{s_{f,z}}{r_{l,k}} \quad (\text{Xie 10})$$

Variables (Xie): $p_{k,f,z} \in \{0, 1\}$ = chỉ báo user k có yêu cầu $v_{f,z}$ hay không;
 $y_{l,f,z} \in \{0, 1\}$ = cache indicator (UAV l , content f , bitrate z); $s_{f,z}$ = chunk size at bitrate z ; $r_{l,k}$ = downlink rate from Eq. (Xie 5).

3.3 Scenario 2 — Transcoding Hit

UAV l has a higher-bitrate version z^+ and transcodes down to z .

$$D_{l,k}^2 = \sum_{f=1}^F \sum_{z=1}^Z p_{k,f,z} (1 - y_{l,f,z}) h_{l,f,z} \left(\frac{s_{f,z}}{r_{l,k}} + \frac{w_0 (s_{f,z^+} - s_{f,z})}{c_{l,k}} \right) \quad (\text{Xie 11})$$

Variables: $h_{l,f,z} = \min(\sum_{z' > z} y_{l,f,z'}, 1)$ = indicator that a higher bitrate is cached; z^+ = lowest cached bitrate above z ; w_0 = CPU cycles per bit for transcoding; $c_{l,k} = C_{\text{comp}}/M$ = computing resource allocated to user k .

3.4 Scenario 3 — Cache Miss

Content not available; UAV fetches from MBS backhaul.

$$D_{l,k}^3 = \sum_{f=1}^F \sum_{z=1}^Z p_{k,f,z} \left(1 - \sum_{z'=z}^Z y_{l,f,z'} \right) \left(\frac{s_{f,z}}{r_{l,k}} + \frac{s_{f,z}}{r_{\text{BS},l}} \right) \quad (\text{Xie 12})$$

$r_{\text{BS},l}$ = backhaul rate from Eq. (Xie 9).

3.5 ABR segment size and client buffer (thesis implementation)

Trong code hiện tại, các representation ABR dùng *cùng* thời lượng segment T_{seg} (giây) như MPEG-DASH; kích thước chunk (bit) thống nhất với Eq. (Xie 10)–(Xie 12) là

$$s_{f,z} = R_z T_{\text{seg}}, \quad (\text{impl-ABR-s})$$

với R_z bitrate nominal của representation z (bit/s). Tham số `abr_segment_duration_s`, `abr_bitrate_values_kbps` trong `config.py`; hàm `models._chunk_size_bits` và `models.abr_bitrate_kbps_list` đảm bảo cùng bảng R_z giữa độ trễ và cache.

Gọi D là độ trễ phục vụ một request theo một trong ba kịch bản (Xie 10), (Xie 11) hoặc (Xie 12) — `calculate_total_cost` trong `models.py`. Buffer phía client (giây):

$$B_{t+1} = \min(B_{\text{max}}, \max(0, B_t + T_{\text{seg}} - D)). \quad (\text{impl-buffer})$$

Thời gian rebuffering trong bước là $\max(0, -(B_t + T_{\text{seg}} - D))$ khi $B_t + T_{\text{seg}} < D$.

3.6 Total System Delay Objective

$$D_{\text{tot}} = \sum_{l \in \mathcal{L}} \sum_{k \in \mathcal{K}} x_{l,k} \cdot D_{l,k} \quad (\text{Xie 14})$$

Variables: $x_{l,k} \in \{0, 1\}$ = user association indicator (user k assigned to UAV l); $\mathcal{L}$ = set of UAVs; $\mathcal{K}$ = set of users.

4 Optimization Problem

4.1 Joint CA+UA Optimization (Xie et al.)

$$\min_{\mathbf{x}, \mathbf{y}} \quad D_{\text{tot}} = \sum_{l \in \mathcal{L}} \sum_{k \in \mathcal{K}} x_{l,k} \cdot D_{l,k} \quad (\text{Xie P})$$

$$\text{s.t.} \quad \sum_{l \in \mathcal{L}} x_{l,k} = 1, \quad \forall k \in \mathcal{K} \quad (15b)$$

$$\sum_{k \in \mathcal{K}} x_{l,k} \leq M, \quad \forall l \in \mathcal{L} \quad (15c)$$

$$\sum_{f,z} y_{l,f,z} \cdot s_{f,z} \leq C_l^{\text{cache}}, \quad \forall l \quad (15a)$$

$$x_{l,k}, y_{l,f,z} \in \{0, 1\} \quad (15d)$$

Ghi chú: Luận văn dùng D3QN để học chính sách; hàm mục tiêu lý thuyết trong [1] là tối thiểu hóa độ trễ D_{tot} ; phần thưởng thực tế là QoE ABR (Mục 5.5).

5 D3QN và cài đặt luận văn

5.1 Q-Value Update (Double DQN)

$$Q_1(s_t, a_t) \leftarrow Q_1(s_t, a_t) + \alpha \left(r_{t+1} + \gamma Q_2 \left(s_{t+1}, \arg \max_{a'} Q_1(s_{t+1}, a') \right) - Q_1(s_t, a_t) \right) \quad (\text{D3QN-a})$$

$$Q_2(s_t, a_t) \leftarrow Q_2(s_t, a_t) + \alpha \left(r_{t+1} + \gamma Q_1 \left(s_{t+1}, \arg \max_{a'} Q_2(s_{t+1}, a') \right) - Q_2(s_t, a_t) \right) \quad (\text{D3QN-b})$$

5.2 Dueling Network Q-Value Decomposition

$$Q(s, a) = V(s) + \left(A(s, a) - \frac{1}{|\mathcal{A}|} \sum_{a'} A(s, a') \right) \quad (\text{D3QN-duel})$$

Variables: $V(s)$ = state value function; $A(s, a)$ = advantage function; normalization term ensures identifiability.

5.3 Action Selection

$$a_t = \arg \max_a \left(V(s_t) + A(s_t, a) - \frac{1}{|\mathcal{A}|} \sum_{a'} A(s_t, a') \right) \quad (\text{D3QN-act})$$

5.4 Target Value (D3QN)

$$Q_{\text{target}} = R_{t+1} + \gamma \max_A Q(S_{t+1}, A; \theta_t^-) \quad (\text{D3QN-tgt})$$

5.5 Reward trong code (QoE ABR)

`environment.py` dùng phần thưởng QoE kiểu Pensieve cho ABR (gắn với độ trễ D từ mô hình Xie ở trên):

$$r_t = u(R_{z_t}) - \beta_{\text{rebuf}} t_{\text{rebuf},t} - \beta_{\text{sw}} |R_{z_t} - R_{z_{t-1}}|_{\text{norm}}, \quad (\text{impl-QoE})$$

trong đó $u(\cdot)$ là log hoặc tuyến tính theo bitrate nominal R_{z_t} ; $t_{\text{rebuf},t}$ lấy từ cập nhật buffer (Mục 3.5); β_{rebuf} , β_{sw} là `abr_rebuffer_penalty`, `abr_switch_penalty` trong `config.py`.

5.6 Vector trạng thái trong code

Vector trạng thái có số chiều cố định theo topology:

$$|\mathbf{s}| = 2 + 7L + 2Z + 3, \quad (\text{impl-state})$$

với L số UAV, $Z = |\{\text{bitrate levels}\}|$: vị trí xe và UAV (chuẩn hóa), khoảng cách xe-UAV, độ đầy cache từng UAV, ba đặc trưng/UAV (cache hit tại (f, z_{req}) , tải, khả năng transcode), trọng số joint Zipf $P(f_{\text{req}}, z_{\text{req}})$ (không phải chỉ báo $p_{k,f,z}$ của [1]), one-hot z_{req} , buffer chuẩn hóa, one-hot bitrate bước trước, throughput EWMA chuẩn hóa. Chi tiết: `VanetEnvironment.get_state` trong `environment.py`.

5.7 Không gian hành động trong code (Xie, UAV-only)

Cài đặt luận văn (tầng UAV, Xie): $a \in \{0, \dots, LZ \cdot 2 - 1\}$ với giải mã (`uav_idx`, `z_cached`, `cache_dec`); chỉ UAV (không MBS trong action); xem `encode_action` / `_decode_action` trong `environment.py`.

6 QEA Baseline (for Comparison)

6.1 Q-bit Observation

$$x_{l,k} = \begin{cases} 1 & \text{if rand() < } |\beta_{l,k}|^2 \\ 0 & \text{otherwise} \end{cases} \quad (\text{Xie 21})$$

$$y_{l,f,z} = \begin{cases} 1 & \text{if rand() < } |\beta_{l,f,z}|^2 \\ 0 & \text{otherwise} \end{cases} \quad (\text{Xie 22})$$

6.2 Rotation Gate Update

$$\begin{pmatrix} \alpha' \\ \beta' \end{pmatrix} = \begin{pmatrix} \cos \Delta\theta & -\sin \Delta\theta \\ \sin \Delta\theta & \cos \Delta\theta \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \quad (\text{Xie 23-24})$$

$\Delta\theta$ is determined by the lookup Table 1 in [1] based on $(x_{\text{cur}}, b_{\text{best}}, f(\mathbf{X}_i) \geq f(\mathbf{B}))$.

Symbol Table

References

- [1] Y. Xie et al., “Joint Caching and User Association Optimization for Adaptive Bitrate Video Streaming in UAV-Assisted Cellular Networks,” *IEEE Access*, 2022.

Table 1: Main Notation Summary (Xie + thesis implementation)

Symbol	Definition	Source
$r_{l,k}$	Downlink rate UAV l – user k	Xie 5
$r_{BS,l}$	Backhaul rate MBS – UAV l	Xie 9
$g_{l,k}$	Average path loss (dB)	Xie 1–2
$P_{l,k}^{\text{LoS}}$	LoS probability	Xie 3
$p_{k,f,z}$	Theo Xie: chỉ báo $\{0,1\}$. Trong mô phỏng: joint Zipf $P(f,z)$; vector trạng thái nhúng $P(f_{\text{req}}, z_{\text{req}})$.	Xie (10–12)
$D_{l,k}^1$	Direct hit delay	Xie 10
$D_{l,k}^2$	Transcoding hit delay	Xie 11
$D_{l,k}^3$	Cache miss delay	Xie 12
D_{tot}	Total system delay objective	Xie 14
$s_{f,z}$	Chunk size (bit); impl.: $R_z T_{\text{seg}}$	impl-ABR-s
R_z	Nominal bitrate, representation z	config
T_{seg}	ABR segment duration (s)	config
$x_{l,k}$	User association indicator	Xie 15
$y_{l,f,z}$	Cache placement indicator	Xie 15
$Q(s,a)$	Q-value (Dueling DQN)	D3QN-duel
r_t (thesis)	QoE ABR reward	impl-QoE